

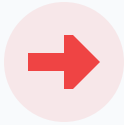
KnowSelf: Agentic Knowledgeable Self-Awareness for LLM Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

"Enabling agents to autonomously regulate knowledge utilization"
"through situational self-awareness"

Current Agent Training Uses "Flood Irrigation"

Indiscriminate knowledge injection hurts efficiency and causes pattern collapse



Trajectory Imitation

e.g., ReAct

Blindly imitates gold trajectories regardless of whether the agent actually needs guidance at a step.

Know%: 0% (no knowledge)

Fragile on unexpected inputs



Trial-and-Error

e.g., Reflexion

Relies on external feedback loops for every step, even when the agent could act correctly alone.

Know%: 0% (reflection only)

High inference overhead



Knowledge-Augmented

e.g., KnowAgent, WKM, ExpeL

Injects domain knowledge at every step, even when the agent already knows the correct action.

Know%: 100% (always injected)

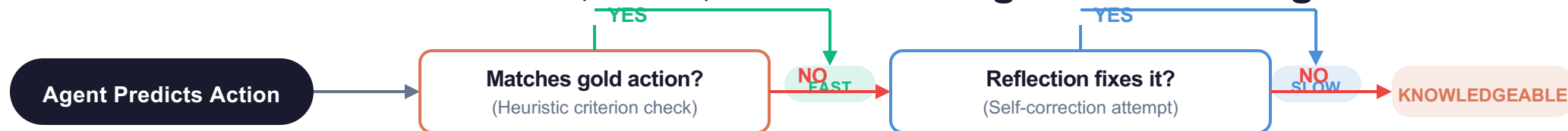
Wasteful, can hurt performance



Key Insight: All three paradigms resemble unconscious pattern-fitting — "flood irrigation" without situational awareness.

Humans possess metacognitive self-awareness: knowing when to act, reflect, or seek help. Agents should too.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking



Fast Thinking

Direct correct action

Decision Criterion:

Predicted action matches gold action

Behavior:

Agent acts directly with minimal deliberation — no extra overhead

Special Token:

<FAST>

Triggers: Direct action execution



Slow Thinking

Self-correct via reflection

Decision Criterion:

Initial action wrong, but reflection produces the correct action

Behavior:

Agent pauses, reflects on error, and self-corrects without help

Special Token:

<REFLECTION>

Triggers: Internal reflection loop



Knowledgeable

Consult external knowledge

Decision Criterion:

Cannot produce correct action even after reflection

Behavior:

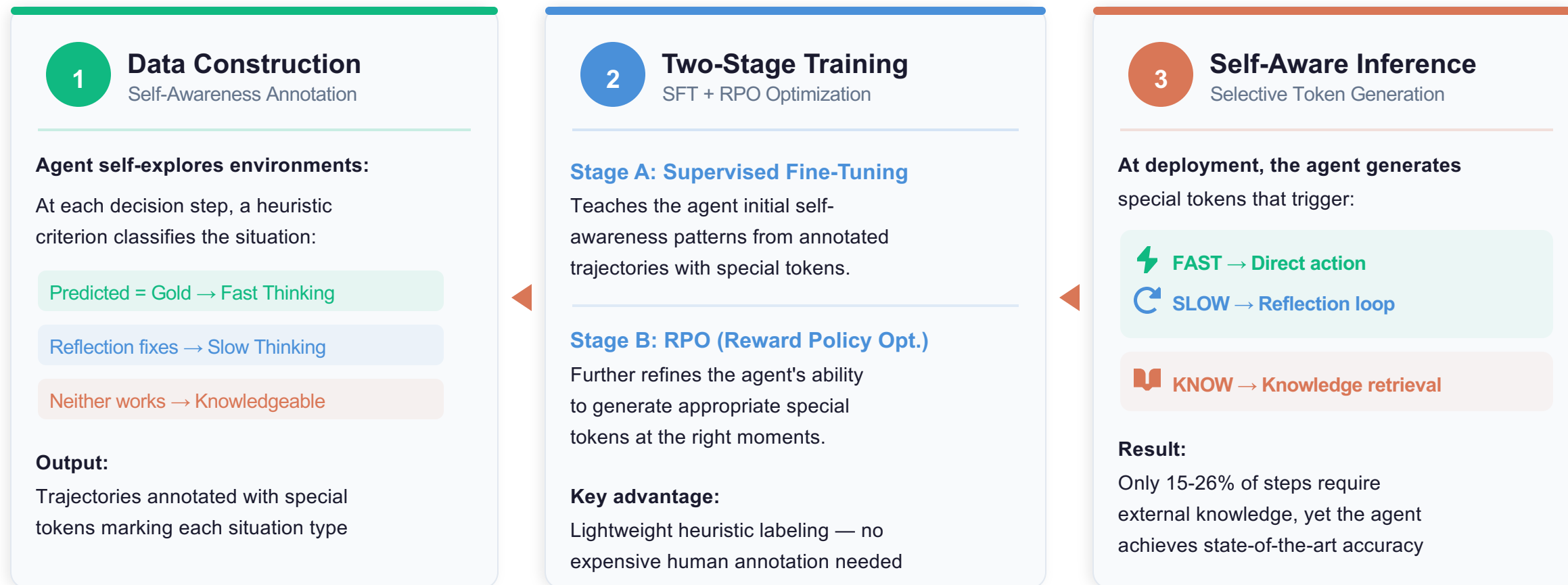
Agent retrieves from external knowledge base to guide action

Special Token:

<KNOWLEDGE>

Triggers: Knowledge base retrieval

KnowSelf: Data Construction → Two-Stage Training → Self-Aware Inference



KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: "Put a hot mug in cabinet"

OpenAI-O1

FAILS

After opening the microwave, incorrectly tries to take an egg from the microwave.

Does not realize it should heat the mug using the already-open microwave.

KnowSelf (Llama-8B)

SUCCEEDS

Uses a Reflection token to realize it should heat the mug using the open microwave, then places it in the cabinet.

<REFLECTION>

Task: "Put a saltshaker in drawer"

DeepSeek-R1

FAILS

Reflects that the drawer is open and tries to close it first, which fails and blocks further progress.

Overthinks — closes drawer unnecessarily.

KnowSelf (Llama-8B)

SUCCEEDS

Applies a Knowledge token noting that the agent should place the object without removing unrelated items from the drawer.

<KNOWLEDGE>



KnowSelf selects the right cognitive mode per step — reflection when needed, knowledge when stuck, fast action when confident.

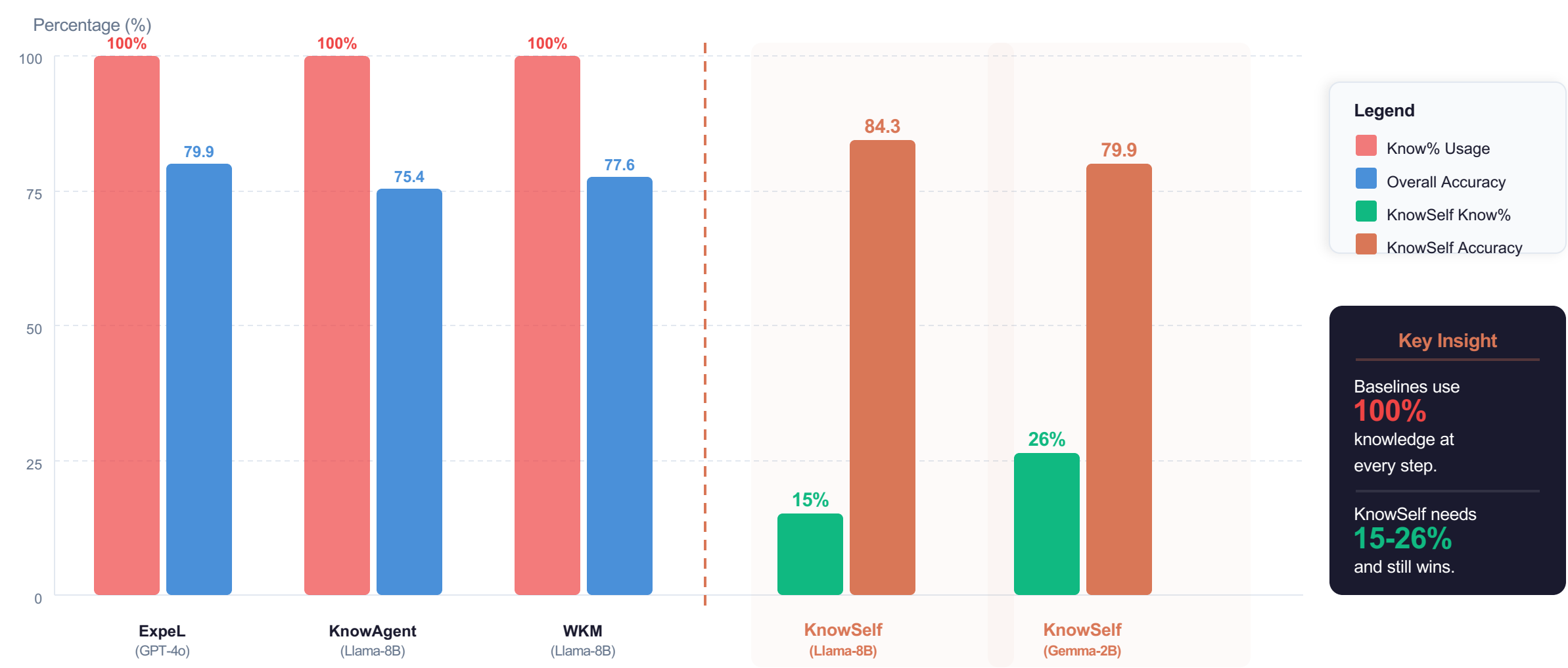
EMPIRICAL RESULTS

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o									
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B									
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B									
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) on ALFWorld. Green Know% = selective use.Red 100% = indiscriminate injection. KnowSelf rows highlighted in orange.

Minimal Knowledge, Maximum Impact: 15-26% Usage Beats 100% Injection



Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful and suboptimal

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform selective approaches

2

Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories — no expensive human annotation required

3

KnowSelf (Llama-8B) surpasses GPT-4o baselines on ALFWorld

84.33% vs. GPT-4o ReAct 76.12% and GPT-4o ExpeL 79.85% — a smaller model beating a frontier model

4

Only 15-26% knowledge usage vs. 100% for competing methods

Massive efficiency gain — selective knowledge retrieval outperforms blanket application at every scale

5

The paradigm generalizes across model scales

Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL — self-awareness scales down effectively

**Future Directions:**

Multi-domain generalization, dynamic knowledge base integration, extending to multi-agent coordination